We know,

No. of coils C = No.of commutator segments

Hence, C = 6

Back pitch, $Y_b = \frac{2C}{P} \pm K$

$= \frac{2X6}{2} \pm K$

= 6 + 1 = 7 (or)

6-1 = 5

Coil pitch, Y = ±2m

m = 1 (for simplex)

Then, Y= 2 X 1 =2

Front pitch, $Y_f = Y_b - Y$

= 7 – 2 = 5 (or)

5 - 2 = 3

Selected Y_b and Y_f are

$Y_b = 7$

$Y_f = 5$

Winding table

Sl. No.	$Y_b = 7$	$Y_f = 5$
1	1	8
2	3	10
3	5	12
4	7	14(2)
5	9	16(4)
6	11	18(6)

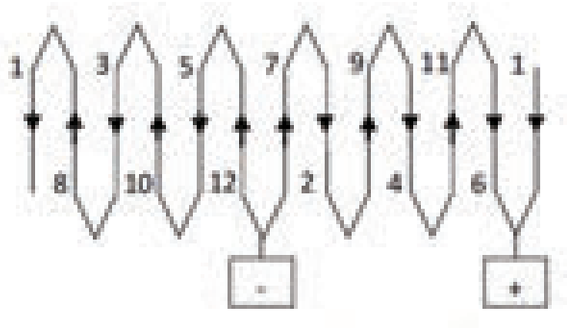


Fig. 9.9 Winding-commutator connection

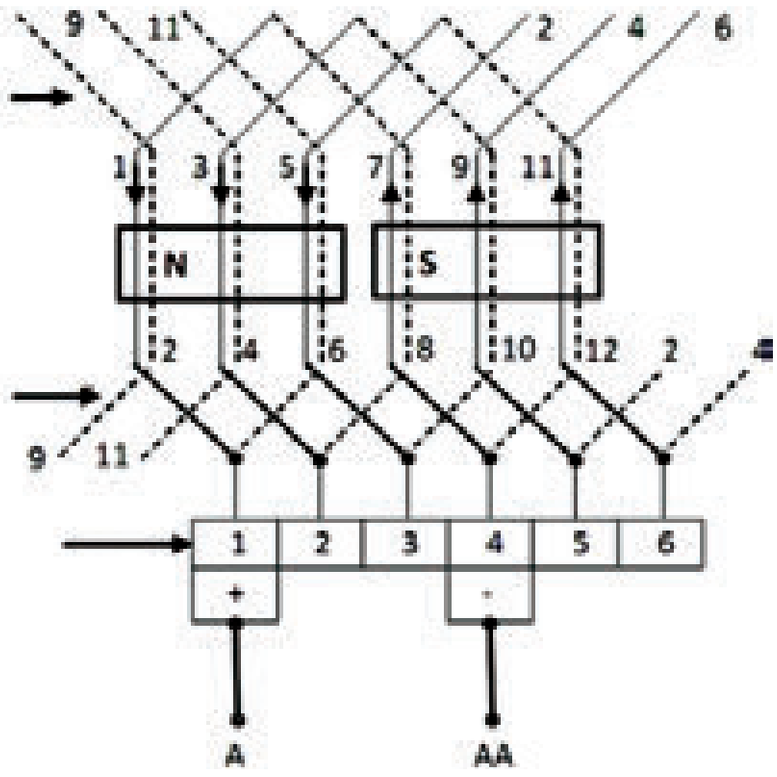


Fig. 9.10 2 Poles 6 slot simplex lap winding diagram

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